Bhavesh Bakshi

Robotics Systems Engineer | Humanoids, Reinforcement Learning, Sim-to-Real Deployment

bhaveshbakshi633@gmail.com +91 88250 95037 Gurgaon, India linkedin.com/in/bhavesh-bakshi633
github.com/bhaveshbakshi633 Portfolio

EXPERIENCE

Mechatronics Engineer
Oct 2022 - Present

SS Innovations - Surgical Robotics | Gurgaon, India

- Architected and deployed **29-DOF bipedal humanoid** system integrating whole-body control, SLAM navigation, voice AI, and RL-trained locomotion on production hardware
- Built end-to-end sim-to-real pipeline using Isaac Gym with **4,096 parallel environments**; achieved **<5% performance gap** on physical robot deployment
- Developed XR teleoperation system (Quest 3) with **<50ms end-to-end latency** enabling real-time dexterous manipulation via inverse kinematics
- Implemented voice AI pipeline: Whisper STT + Ollama LLM with RAG + neural TTS; **200ms** voice-to-response, **50ms** emergency stop
- Engineered **500Hz** real-time control loop coordinating 14-DOF arms, 12-DOF legs, and 3-DOF waist via Unitree SDK2 (CycloneDDS)
- Led system integration for surgical robotics platforms: motion control, encoder integration, safety systems

Lead Technical Consultant
Apr 2022 - Sep 2022

Learning Links Foundation - EdTech NGO | India

- Trained **1,000+ students** across Kendriya Vidyalayas in robotics, IoT, and embedded systems
- Mentored **30+ projects** from concept to working prototype; developed hands-on curriculum for practical learning

Engineering Intern
Jun 2019 - Aug 2019

Heavy Engineering Corporation Ltd | Ranchi, India

- Assisted project planning and execution for heavy machinery manufacturing; participated in cross-functional engineering reviews

TECHNICAL SKILLS

Robotics & Control: ROS2, Unitree SDK2, MuJoCo, Isaac Gym, SLAM, IK/FK

ML/RL: PyTorch, Stable-Baselines3, PPO, ONNX Runtime, TensorBoard

CAD/FEA: SolidWorks, Fusion 360, Ansys Workbench, Creo

Infrastructure: Docker, Git, Linux, CycloneDDS, WebSocket, REST

Programming: Python, C++, MATLAB, Bash

Vision: OpenCV, RealSense SDK, ArUco, MediaPipe, Point Clouds

Embedded: Arduino, Raspberry Pi, ESP32, EtherCAT, CAN

Prototyping: 3D Printing, TIG Welding, PCB Design, Machining

SELECTED PROJECTS

Naamika: Voice-Controlled Humanoid

Full-stack 29-DOF bipedal system with WBC, SLAM, LLM-based voice control, and 22-action safety gatekeeper. Production-deployed.

ROS2, CycloneDDS, Whisper, Ollama, FAISS

ASAP Bipedal Motion Learning

Sim-to-real locomotion via PPO in 4,096 parallel Isaac Gym envs. Deployed walking, kicking, and gesture policies on hardware.

Isaac Gym, PyTorch, ONNX, Domain Randomization

Universal RL Training Center

End-to-end PyQt6 GUI: URDF loading, reward design, curriculum training, domain randomization. 12 integrated modules.

PyQt6, MuJoCo, Stable-Baselines3, ONNX

XR Teleoperation System

Quest 3 hand tracking to robot arms via IK. Sub-50ms latency, 6-DOF mapping, dexterous hand support for manipulation.

OpenXR, Damped LS IK, CycloneDDS, RealSense

Fixed-Wing & Multi-Rotor UAVs

5 RC aircraft iterations + FPV quadcopter. Custom aerodynamic design, PID tuning, successful flight testing.

Betaflight, Arduino, PID Control, FPV

BAJA SAE ATV Chassis (3x)

FEA-optimized 4130 chromoly space frame. 38kg at 2.1+ FOS, 25% cost reduction via DFM optimization.

SolidWorks, Ansys, TIG Welding

EDUCATION

B.E. Mechanical Engineering 2017 - 2021
Birla Institute of Technology, Mesra
CGPA: 7.3/10

Higher Secondary (PCM + CS) 2015 - 2017
Army Public School Damana, Jammu
School Topper: C++, Physics, Chemistry (95/100) | 90%

AWARDS & COMPETITIONS

GOLD World Skills India 2017 - Mobile Robotics (State)
GOLD BIT Patna Robotics 2018
2ND IIT Patna Robotics 2017 - Aqua Soccer
Boeing RC Aircraft Challenge - IIT Kharagpur 2018
GOLD IIT Patna Robotics 2018 - Aqua Soccer
GOLD IIT Delhi Robotryst 2014 - Junior (State)
SAE BAJA India 2019 & 2020
Technoxian-18 National Drone Championship

LANGUAGES

English - Professional **Hindi** - Native **French** - Basic